Local Bayesian-Network Factor Adaptation for Fairness Transfer across States and Years

ICT 5143: Algorithms Fairness and Graphical Models Laboratory
Problem 5; Team 5
Aslam Mugutkhan, 261090830010; Rao Ashlesha Padmananda, 261090830019
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Abstract

Can updating at most two shifted Bayesian-network factors preserve equalized odds better than global adaptation when only 200 or 1,000 target labels are available? On ACSIncome (CA, TX, NY; 2018–2019; nine transfers) we propose BN-LFA: Bayes-factor × decision-impact factor selection, then equalized-odds thresholds refitted on source data reweighted by the selected factors. BN-LFA was fairer than target threshold adjustment (mean Δ_{EOdds} -0.039 and -0.065, Holm $p = 0.006$) and than updating every factor at $n = 200$ (-0.014, $p = 0.032$), but not than covariate reweighting, and not clearly than keeping the source rule.

1 Question, context and contribution

A classifier made fair in one state or year is often deployed in another; fairness under distribution shift is established research [1], with transferability bounds [2] and guarantees under demographic shift [3]. Task: Folktables ACSIncome [4], $Y = 1$ if annual income exceeds \$50,000; $D = 1$ is a high-income prediction (a benchmark classification, not merit or need). Stakeholders are the people in the census data and the developers and auditors of transferred models. A Bayesian network (BN) localises a shift to the factors whose tables change (the domain S becomes their extra parent [5]).

Research question: with $n \in \{200, 1000\}$ labelled target persons, does updating ≤ 2 factors give a lower target equalized-odds gap than updating all factors, target threshold adjustment and covariate reweighting, losing at most 0.02 balanced accuracy (BA)?

Contribution: BN-LFA, a factor-local rule whose thresholds use only the density ratio of the updated factors; refuted for a comparator if the Holm-adjusted paired household-bootstrap test shows no reduction.

Closest work: Schrouff et al. [6] diagnose fairness-transfer failures with a graph but do not repair them. We reproduce covariate-shift correction [7, 8, 9] as IW-EO and threshold post-processing [10] as EO-S/TT. Fairness constraints in training [11], graph-based transport of predictors [12, 13] and target- or label-shift correction [14, 15] address related problems without a label-budgeted, factor-local repair.

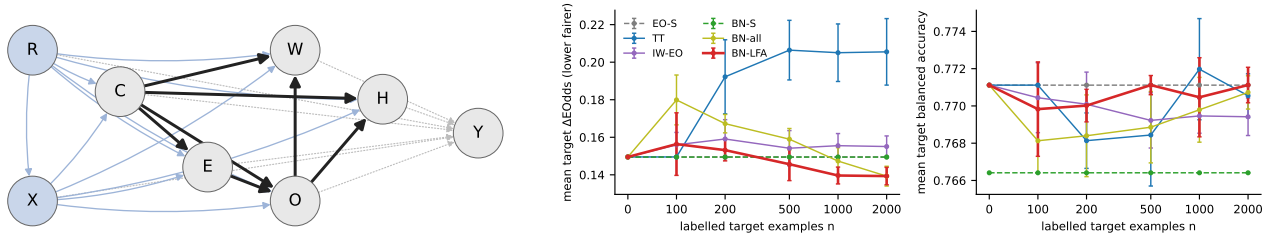
2 Data, graphical model and fairness method

Data and estimand: ACS PUMS 1-Year person files [16], CA, TX, NY, 2018 and 2019, filtered exactly as ACSIncome (verified row by row with `folktables`); 103k–197k persons per domain, split by household into train/validation/test (60/20/20 %). Groups: 18 recorded race × sex cells; primary gaps use groups with ≥ 100 positives and negatives in a test split (7–10 per domain), smaller cells are exploratory. Estimand: the unweighted benchmark sample (survey-weighted estimates secondary).

Graph and inference: race R, sex X, age C, education E, class of worker W, occupation O, hours H and Y, all observed. Hill climbing on validation likelihood (tiers, required R, X parents, in-degree ≤ 4) learned the same associational graph in all six domains (Fig. 1a):

$$P(V) = P(R)\,P(X \mid R) \prod_{v \in \{C,E,W,O,H\}} P(v \mid R, X, \mathrm{pa}'(v))\; P(Y \mid R, X, C, E, W, O, H),$$

$\mathrm{pa}'(E) = \{C\}$, $\mathrm{pa}'(O) = \{C, E\}$, $\mathrm{pa}'(W) = \mathrm{pa}'(H) = \{C, O\}$; smoothed Dirichlet CPTs and a log-linear outcome CPD tabulated over 57,600 configurations. The audit query $P(G{=}g, \mathrm{pa}(Y), Y)$ gives the model-implied group TPR/FPR of any rule; variable elimination (four orders), a clique tree [17], enumeration and pgmpy agree to 1.3×10^{-15} (tolerance 10^{-9}).



(a) learned graph; blue: required R, X edges

(b) mean of 9 transfers; bars: s.d. over 5 seeds

Figure 1: (a) Structure learned in all six domains. (b) Target Δ_{EOdds} and balanced accuracy against the label budget.

Method and fairness: BN-LFA scores every factor by a Bayes factor of "shifted" (averaged over tilt and Dirichlet families) and by the exact change of the source rule's gap and BA if only that factor is updated, updates the ≤ 2 factors with $\log \text{BF} > 0$ and largest $P(\text{shift}) \times \text{impact}$, and refits thresholds on the source validation split weighted by $w = \prod_{v \in F} \hat{\theta}_t(v \mid \text{pa}) / \theta_s(v \mid \text{pa})$ ($F = \varnothing$ gives the source rule). Primary criterion: equalized odds within each target domain, $\Delta_{\text{EOdds}} = \max(\text{TPR range}, \text{FPR range})$, because errors of a favourable classification should not depend on the group given the true label; demographic parity and calibration are audits. Utility: BA, allowed loss $\delta = 0.02$. Race and sex are used at prediction time (features and group thresholds).

3 Evaluation protocol

Baselines: LR-S and GBT-S (source models), reweighing (RW-S) [18], source EO post-processing (EO-S) [10], target threshold adjustment (TT), importance-weighted EO (IW-EO) [7], target retraining (LR-T, GBT-T), source BN (BN-S) and all factors updated (BN-all). Ablations: random factors, no decision impact (BN-LL), all factors, Dirichlet-only updates. All methods in a run get the same $n \in \{0, 100, 200, 500, 1000, 2000\}$ persons from the target's train+validation pool; the target test split is used only for evaluation (shuffling its labels leaves every decision unchanged). Thresholds come from one shared routine on validation or adaptation data; five seeds. Confirmatory family: BN-LFA vs BN-all, TT, IW-EO at $n = 200$ and 1000 (mean of nine transfers), paired household bootstrap ($B = 2000$, decisions fixed), Holm $\alpha = 0.05$, BA non-inferiority at -0.02. Experiment 4 is a *synthetic* simulation with known shifted factors. Robustness: bins, pooled sources, top factor excluded, unlabelled covariates, survey weights.

4 Results and interpretation

Method	BA ($n = 200$ / 1000)	Δ_{EOdds} ($n = 200$ / 1000)	Log loss / Brier
EO-S (source rule)	0.771 / 0.771	0.150 / 0.150	0.448 / 0.147
TT (target thresholds)	0.768 / 0.772	0.192 / 0.205	0.448 / 0.147
IW-EO (closest baseline)	0.770 / 0.769	0.159 / 0.156	0.449 / 0.147
LR-T (target retraining)	0.738 / 0.763	0.521 / 0.453	0.461 / 0.151
BN-all (ablation)	0.768 / 0.770	0.167 / 0.147	0.445 / 0.146
BN-rand (ablation)	0.771 / 0.770	0.165 / 0.152	0.447 / 0.146
BN-LL (ablation)	0.770 / 0.772	0.150 / 0.143	0.445 / 0.146
BN-LFA (proposed)	0.770 / 0.770	**0.153 / 0.140**	0.445 / 0.146

Table note: means of 9 transfers $\times$ 5 seeds; BA higher, Δ_{EOdds} lower is better; log loss and Brier at $n = 1000$; target test splits of 20,681–39,551 persons; operating point $\delta = 0.02$. Paired BN-LFA minus comparator, household-bootstrap 95 % CI, Holm p: TT -0.039 $[-0.045, -0.030]$ and -0.065 $[-0.070, -0.047]$ ($p = 0.006$); BN-all -0.014 $[-0.020, -0.004]$ ($p = 0.032$) and -0.008 $[-0.013, 0.001]$ ($p = 0.246$); IW-EO -0.006 $[-0.011, 0.003]$ and -0.016 $[-0.022, 0.001]$ ($p \geq 0.22$). BA was non-inferior in all six tests.

Main finding: *partially supported.* At matched utility (BA within 0.003), local adaptation beat spending the labels on target thresholds at both budgets and beat updating every factor at $n = 200$,

but not covariate reweighting; its gain over keeping the source rule was exploratory and not significant (-0.010 $[-0.018, 0.004]$ at $n = 1000$). The source rule degraded little after zero-label transfer (mean $+0.028$, worst $+0.096$), leaving little to repair.

Mechanism and failure: the diagnostic tracks the fairness change (predicted vs. observed change of the source rule's gap over the nine pairs: Spearman 0.88, $p = 0.002$, exploratory). State transfers mostly select education $P(E \mid R, X, C)$ and the outcome factor; race composition is detected first but barely moves group error rates. Dropping decision impact (0.143) or random factors (0.152) weakened BN-LFA (0.140) only slightly. At $n = 200$, updating every factor lowered held-out joint log-likelihood (-8.926 vs. -8.898 per person), BN-LFA raised it (-8.873). In the synthetic study, BN-LFA with unlabelled covariates repaired a group-specific composition shift that IW-EO could not (0.134 vs. 0.220; EO-S 0.241). It failed when three factors shifted (0.187 vs. 0.184 for EO-S), with coarse bins (0.150 vs. 0.145) and when its top factor was excluded ($+0.009$).

5 Limitations, ethics and conclusion

The graph is associational and its tiers are assumptions; results describe the unweighted sample of three states and two years (survey-weighted BN-LFA$-$TT averages -0.072, per-pair replicate SEs up to 0.066). Gaps are max$-$min statistics over 7–10 groups; American Indian, Alaska Native and Pacific Islander cells are too small for any claim. The nominal \$50k label ignores cost of living and encodes historical inequality, and group thresholds use race and sex, which may be unlawful in practice. Conclusion: with few labels, adapting ≤ 2 factors is a safer use of target data than refitting thresholds, but here it offers no reliable gain over reweighting or no adaptation. Next: a third year.

Reproducibility and individual contributions

Reproduce: *Run all* in `problem5_fairness_transfer_bn.ipynb` (Google Colab CPU, 13 min; about 30 min on a 4-core laptop); settings: its `CFG` cell; versions in `results/reproducibility.json`; headline table: `results/metrics/confirmatory_tests_holm.csv`.

Aslam Mugutkhan: [*implementation/analysis artifacts and verification*].

Rao Ashlesha Padmananda: [*implementation/analysis artifacts and verification*].

Evidence: `results/`; requirements matrix and full related-work comparison in `docs/`.

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